
Exercise 7

1 Tube MPC

Consider the following scalar linear discrete-time system

$$x^+ = 1.1x + u + w$$

under the constraints $x \in \mathbb{X} = [-2, 2]$ and $u \in \mathbb{U} = [-0.8, 0.8]$ and subject to a bounded disturbance $w \in \mathbb{W} = [-0.2, 0.1]$. Design a tube-based MPC controller for this system with the following steps:

1. Compute the minimal robust invariant set $\mathcal{E}$ for this system under the control law $u = Kx$ with $K = -0.3$.
Hint: $[a, b] \oplus [c, d] = [a + c, b + d]$ and $\sum_{i=0}^{\infty} x^i = \frac{1}{1-x}$ if $|x| < 1$.
2. Compute the tightened constraints $\mathbb{X} \ominus \mathcal{E}$ and $\mathbb{U} \ominus K\mathcal{E}$.
3. Verify terminal set feasibility. Check if $\mathcal{X}_f = [-1, 1.5]$ ensures recursive feasibility for terminal control law $u = -0.3x$.
4. Write out the formulation of a tube-MPC controller with all ingredients above.